Rae J. Holcomb

NASA Post-doctoral Fellow · Exoplanets & Stellar Astrophysics

raeholcomb15@gmail.com | rae-holcomb.github.io/ | rae-holcomb | @raeleighscatter | 0000-0002-5034-9476

Research Interests

M Dwarf Stars Characterizing and interpreting stellar variability in M dwarfs
Stellar Rotation Techniques for identifying stellar rotation periods in TESS light curves
Exoplanets High-precision radial velocity planet search and characterization

Education & Research Positions

University of California, Irvine — *Irvine, CA*
Ph.D. in Physics and Astronomy — *Aug 2019 - Spring 2025 (expected)*
- Advisor: Dr. Paul Robertson
- Chancellor Fellowship
- Dissertation: *Maximizing the Yield of Space-based Photometry to Characterize Exoplanets and Their Host Stars*

NASA Goddard Space Flight Center — *Greenbelt, MD*
Pandora Science Team Intern — *Oct 2023 - Oct 2024*
- Advisor: Dr. Christina Hedges
- Project: Data Pipeline and Commissioning Software

Center for Computational Astrophysics @ The Flatiron Institute — *New York City, NY*
Pre-Doctoral Fellowship — *Aug 2022 - July 2023*
- Advisor: Dr. Ruth Angus
- Fellowship Project: The TESS Rotation Collaboration

Rice University — *Houston, TX*
B.S. in Astrophysics — *Aug 2015 - June 2019*
- Advisor: Dr. Patrick Hartigan
- Graduated with Distinction in Research and Creative Works
- Senior Thesis: *Identifying Young Fast-Rotators from TESS Light Curves*

Teaching & Work Experience

Code/Astro Teaching Assistant & Mentor — *Evanston, IL*
Code/Astro Software Development Workshop — *Jul 2022 - Jul 2023*
- Code/Astro is a week-long astronomy software development workshop that aims to teach fundamental software engineering skills and best practices for sustainable open-source packages for astronomy applications.
- Assist with teaching curriculum covering git control, software structure, code documentation, package publishing, and more.
- Mentor students in small groups, facilitating an interactive and hands on learning environment.

Instructor of Record, Phys 20A — *Irvine, CA*
UC Irvine — *July 2021*
- Sole instructor for Phys 20A - Introduction to Astronomy summer session classes
- Managed course materials, grading, registration, etc

Teaching Assistant, Various — *Irvine, CA*
UC Irvine — *Sept 2019 - Present*
- 10 quarters of experience teaching various physics and astronomy classes, including lecture and lab style classes
- Duties include leading discussion sections, teaching labs, grading papers, and offering personalized tutoring

Teaching Assistant, ASTR 202 — *Houston, TX*
Rice University — *Jan 2019 - May 2019*
- Teaching assistant for introductory level astronomy class
- Hosted office hours, graded, and offered personalized help to 50 students

Technical Writing Mentor, ENGI 120 — *Houston, TX*
Rice University — *Aug 2016 - Dec 2018*
- Tutored freshmen engineering students in the design process as they planned and prototyped innovative solutions for real world clients
- Individually mentored students to hone technical writing skills, offering criticism and feedback on 15-30 essays per student team each semester

Course Designer and Instructor, COLL 131 — *Houston, TX*

Rice University — *Aug 2017 - May 2018*

- Designed and taught a unique credited college course “COLL 131 — Narratives in Interactive Media” at Rice University
- Facilitated literacy in video games and coached students to engage with games critically as an artistic medium
- Emphasized critical analysis of design and storytelling techniques in video games

Software Development Intern — *Houston, TX*

Spiceworks — *May 2016 - Aug 2016*

- Spiceworks is a leading company that develops web tools and market research for the IT industry
- Implemented features for Cloud Helpdesk web application, including front end, back end, web styling, and databases
- Experience coding in Ruby on Rails, JavaScript, and some SQL

Talks & Posters

Seminars (* denotes invited talks)

July 2024	***Seminar**, Exoplanet Watch Seminars	*Remote*
May 2024	***Seminar**, Goddard Space Flight Center Exoplanet Seminars	*Greenbelt, MD*
May 2024	**Seminar**, University of Maryland Lunch Talks	*Greenbelt, MD*
Mar 2024	***Seminar**, Saddleback Community College Outreach Talks	*Mission Viejo, CA*
Oct 2023	**Seminar**, JPL Exoplanet Group Seminars	*Remote*
Dec 2022	**Seminar**, NRAO Professional Development Seminars	*Remote*
Dec 2022	*** Seminar**, American Museum of Natural History Seminars	*New York City, NY*
Nov 2022	*** Seminar**, Caltech/IPAC Departmental Seminar	*Pasadena, CA*
Sep 2022	*** Seminar**, Columbia University Lunch Talks	*New York City, NY*

Conferences

June 2023	**Poster**, Cool Stars V	*San Diego, CA*
Dec 2023	**Talk**, ExoSoCal	*Pasadena, CA*
Jul 2023	**Poster**, Towards other Earths III	*Porto, Portugal*
Mar 2023	**Poster**, Extreme Precision Racial Velocity V	*Santa Barbara, CA*
Mar 2022	**Talk**, 50 Years of Skumanich Relations Conference	*Boulder, CO*
Jan 2020	**Poster**, American Astronomical Society 235th Meeting	*Honolulu, HI*

Leadership & Outreach

TESS Rotation Collaboration (TRC) — *Various*

Principal Investigator — *November 2022 - Present*

- A multi-institutional collaboration I founded to coordinate efforts on measuring stellar rotation periods with the TESS mission, with members from over a dozen institutions.
- I am organizing a community data challenge to facilitate cross-comparison of rotation-finding methods, with results expected in 2025.

Astrobiology Book Club — *Remote*

Founder — *Aug 2022 - Present*

- Organize and run weekly meetings to read and discuss key papers in astrobiology-related fields
- Facilitate research discussions among graduate students across multiple universities

Underrepresented Genders in Physics and Astronomy (UNITY) — *UC Irvine*

Mentor — *Sep 2019 - Present*

- Individually mentor undergraduate physics students from underrepresented demographics
- Help students find research opportunities, develop professional skills, and build confidence
- Organize and lead informational events and panels on topics including applying to graduate schools and finding REU opportunities

Physics & Astronomy Community Excellence (PACE) — *Rice University*

Graduate Peer Mentor — *Sept 2020 - June 2023*

- Peer mentorship program for incoming first year graduate students
- Includes one-on-one mentorship and weekly group meetings on career and community development

Rice Players Astronomy Outreach — *Rice University*

Dramaturg — *Jan 2019 - April 2019*

- Served as the ”astronomy consultant” for a university production of *Silent Sky*, a play about the history of women in astronomy
- Hosted Q&A sessions after each performance to answer audience questions and share knowledge about the science behind the play
- Worked with actors and production members to support accurate depictions of scientific equipment and concepts
- Gave astronomy outreach talks to local high schools who were also putting on performances of the same play

Honors & Awards

2022	**Flatiron Institute Predoctoral Fellowship**,	*New York City, NY*
2020	**UC Grad Slam**, Semifinalist	*Irvine, CA*
2019	**Chancellor's Fellowship**, UC Irvine	*Irvine, CA*
2019	**Distinction in Research and Creative Works**, Rice University	*Houston, TX*
2019	**Dessler Award for Undergraduate Achievement in Astronomy**, Rice University	*Houston, TX*
2019	**Catalyst Science Communication Symposium**, 3rd place	*Houston, TX*
2017	**Dessler Scholarship for Summer Undergraduate Research**,	*Houston, TX*
2016	**Best Interdisciplinary Team Award**, Rice Engineering Design Showcase	*Houston, TX*

Selected Publications

A full list of publications can be found here.

References

[1] **Holcomb, Rae J.**, Paul Robertson, Patrick Hartigan, Ryan J. Oelkers, and Caleb Robinson. SpinSpotter : An Automated Algorithm for Identifying Stellar Rotation Periods with Autocorrelation Analysis. ApJ, 936(2):138, September 2022.

[2] Christina Hedges, **Holcomb, Rae**, Benjamin Hord, Thomas Barclay, Jessie Dotson, and Elisa Quintana. Open-source simulation tools for verifying NASA Pandora SmallSat's scientific performance. In Jorge Ibsen and Gianluca Chiozzi, editors, *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, volume 13101 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 131010F, July 2024.

[3] Jack Lubin, Judah Van Zandt, **Holcomb, Rae**, Lauren M. Weiss, Erik A. Petigura, Paul Robertson, Joseph M. Akana Murphy, Nicholas Scarsdale, Konstantin Batygin, Alex S. Polanski, Natalie M. Batalha, Ian J. M. Crossfield, Courtney Dressing, Benjamin Fulton, Andrew W. Howard, Daniel Huber, Howard Isaacson, Stephen R. Kane, Arpita Roy, Corey Beard, Sarah Blunt, Ashley Chontos, Fei Dai, Paul A. Dalba, Kaz Gary, Steven Giacalone, Michelle L. Hill, Andrew Mayo, Teo Močnik, Molly R. Kosiarek, Malena Rice, Ryan A. Rubenzahl, David W. Latham, S. Seager, Joshua N. Winn, and Kaz Gary. TESS-Keck Survey. IX. Masses of Three Sub-Neptunes Orbiting HD 191939 and the Discovery of a Warm Jovian plus a Distant Substellar Companion. AJ, 163(2):101, February 2022.

[4] P. Hartigan, **Holcomb, R.**, and A. Frank. Proper Motions and Shock Wave Dynamics in the HH 7-11 Stellar Jet. ApJ, 876(2):147, May 2019.

[5] Marc Hon, Daniel Huber, Nicholas Z. Rui, Jim Fuller, Dimitri Veras, James S. Kuszlewicz, Oleg Kochukhov, Amalie Stokholm, Jakob Lysgaard Rørsted, Mutlu Yıldız, Zeynep Çelik Orhan, Sibel Örtel, Chen Jiang, Daniel R. Hey, Howard Isaacson, Jingwen Zhang, Mathieu Vrard, Keivan G. Stassun, Benjamin J. Shappee, Jamie Tayar, Zachary R. Claytor, Corey Beard, Timothy R. Bedding, Casey Brinkman, Tiago L. Campante, William J. Chaplin, Ashley Chontos, Steven Giacalone, **Holcomb, Rae**, Andrew W. Howard, Jack Lubin, Mason MacDougall, Benjamin T. Montet, Joseph M. A. Murphy, Joel Ong, Daria Pidhorodetska, Alex S. Polanski, Malena Rice, Dennis Stello, Dakotah Tyler, Judah Van Zandt, and Lauren M. Weiss. A close-in giant planet escapes engulfment by its star. Nature, 618(7967):917–920, June 2023.

[6] Guðmundur Stefánsson, Suvrath Mahadevan, Yamila Miguel, Paul Robertson, Megan Delamer, Shubham Kanodia, Caleb I. Cañas, Joshua N. Winn, Joe P. Ninan, Ryan C. Terrien, **Holcomb, Rae**, Eric B. Ford, Brianna Zawadzki, Brendan P. Bowler, Chad F. Bender, William D. Cochran, Scott Diddams, Michael Endl, Connor Fredrick, Samuel Halverson, Fred Hearty, Gary J. Hill, Andrea S. J. Lin, Andrew J. Metcalf, Andrew Monson, Lawrence Ramsey, Arpita Roy, Christian Schwab, Jason T. Wright, and Gregory Zeimann. A Neptune-mass exoplanet in close orbit around a very low-mass star challenges formation models. *Science*, 382(6674):1031–1035, December 2023.

[7] Michael Endl, Paul Robertson, William D. Cochran, Phillip J. MacQueen, Brendan P. Bowler, Kyle E. Franson, **Holcomb, Rae**, Corey Beard, Howard Isaacson, Andrew W. Howard, and Jack Lubin. A Jupiter Analog Orbiting The Nearby M Dwarf GJ 463. AJ, 164(6):238, December 2022.